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ASSIGNMENT 1

BRIEF HISTORY OF HYDRAULICS

Hydraulics is a very ancient science. Throughout history, people have always drifted toward the water. The Egyptians and Babylonians constructed canals, both for irrigation and for defensive purposes. Hydraulics, or the use of fluid power to run machinery, have been around for quite a while, for over 2000 years.

The first notable attempts to rationalize the nature of pressure and flow pattern were undertaken by the Greeks. The Greek word “hydraulics” from which the word “hydraulic” comes referred to water (hydra) and pipe (aulos). The Greek developed complex systems of water and hydraulic power, including irrigation systems, canals and aqueducts (Aqueducts provided a way for cities to get reliable supply of water from nearby source and carry it to them for easy access.).

The laws of hydrostatic and buoyancy were enunciated, Ctesibius of Alexandria and Hero of Alexandria designed hydraulic equipment such as the piston pump and water clock and the Archimedes screw pump. The Roman appear, like the Egyptians, to have been more interested in practical and constructional aspects of hydraulics than in theorizing. The Romans had water mills and developed “hushing” an early version of hydraulic mining, for the use of gold fields in the region.

The development continued slowly until the time of Renaissance, when men such as Leonardo Da Vinci began to publish the results of their observations. Ideas which emerged then, respecting conversation of mass (continuity of flow), frictional resistance and the velocity of the surface waves, are still in use, though sometimes in a refined form.

At the start of the 17th century, the study of hydraulics advanced with Simon Stevin’s discovery of the hydrostatic paradox and the study of non-moving water, and Galileo Galilei’s observations on gravity. Galileo studied gravitational acceleration, which played into the movement of water.

The Frenchman Blaise Pascal studied hydraulic fluids. Pascal came up with his law in 1648. Pascal Law says that if pressure is applied to a fluid inside of closed system, the pressure will be equally transmitted in all directions. This principle laid the groundwork for some of the most critical hydraulic innovations in the industrial work. Pascal’s work led to the invention of hydraulic press and provided an extremely valuable principle for anything that works with fluid movement.

About a 150 years later, Joseph Bramah patented the hydraulic press in 1795. In this invention, he used Pascal's discovery to multiple a small amount of force and create large amounts of pressure capable of powering machinery and lifting very heavy objects.

In 1738, Daniel Bernoulli published his book Hydrodynamica, in which he described how water behaves under certain conditions and how it reacts to equilibrium, pressure and speed. Bernoulli's principle was based on his studies of energy conversation. Bernoulli worked closely with the mathematician Leonhard Euler, who may have derived Bernoulli equation first. He also developed several important hydraulic equations and a hydraulic turbine.

The 19th century led to many advancements in hydraulics: the effects of temperature on pipe flow and determining flow rate; the development of the rotary engine and hydraulic accumulator; the introduction and regulations of hydraulic mining.

As the 20th century washed in, so did new, varied uses of hydraulics. Modern methods for estimating flows in pipe and channels. The advent of the electronic computer, and advances in sensing and data logging equipment have revolutionized many aspects of hydraulics.

BRIEF HISTORY OF PNEUMATICS

The origin of pneumatics can be traced back to the first century when ancient Greek mathematician Hero of Alexandria wrote about his inventions powered by steams or the wind. In 1600s, a German physicist Otto Von Guericke invented the first vacuum pump, this made use of pressurized air.

In the 1800s, pneumatics evolved into its own industry with businesses and inventors recognizing its great potential. Innovation began with the introduction of pneumatic tubes in Victorian England that used pipelines to transport telegrams from one station to another.

John Wanamaker introduced pneumatic tubes to the United States Post Office in 1875 to transport letters throughout the buildings. The most elaborate use of these tubes was introduced in 1867 by Alfred Beach. He created a subway line that transported passengers through pipe. Unfortunately, his project was not granted commercial access and closed down after a while.

Since the invention boom of the 1800s , there has been a dramatic increase in the amount and usage of compressed air devices. After the 1960s, many of these pneumatics components were also digitally controlled. Pneumatics are almost everywhere today, from the axial flow and centrifugal compressors in jet engines to the air tool that helps clean teeth at a dentist's office.